C. M. HERSHEY.
COMPTOMETER.
APPLICATION FILED NOV. 17, 1916.

1,280,466

Patented Oct. 1, 1918.
2 SHEETS—SHEET 1.

Fig. 1.

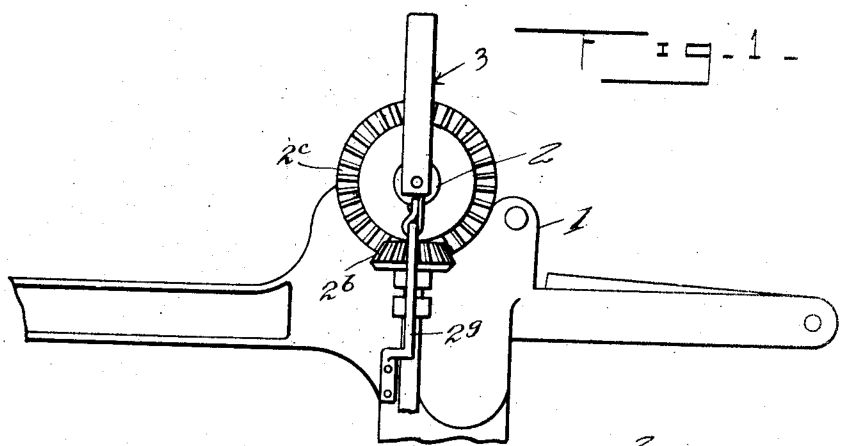



Fig. 6.

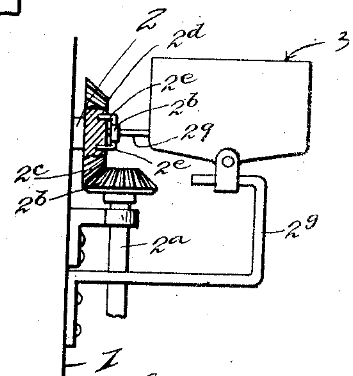
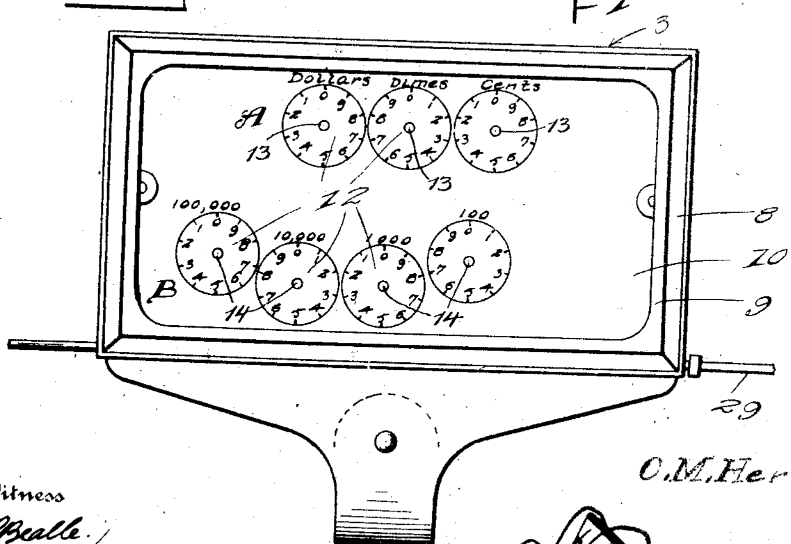



Fig. 2.

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2 SHEETS—SHEET 2.

Fig. 3.

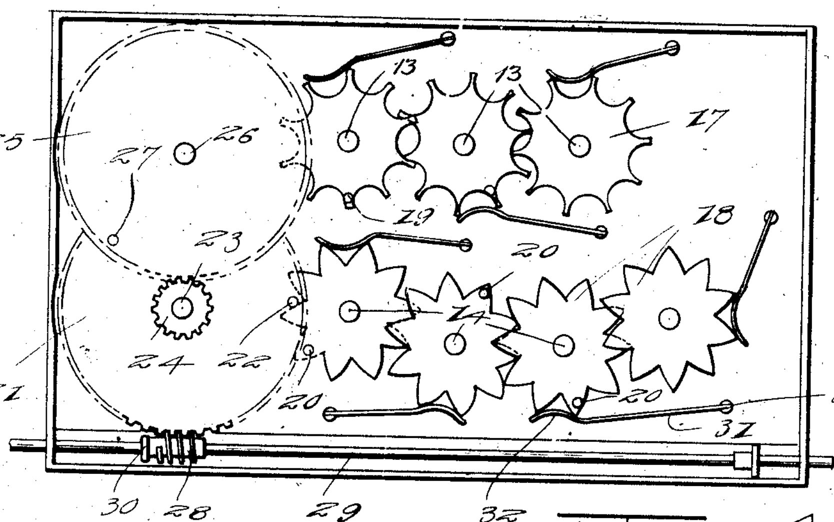



Fig. 4.

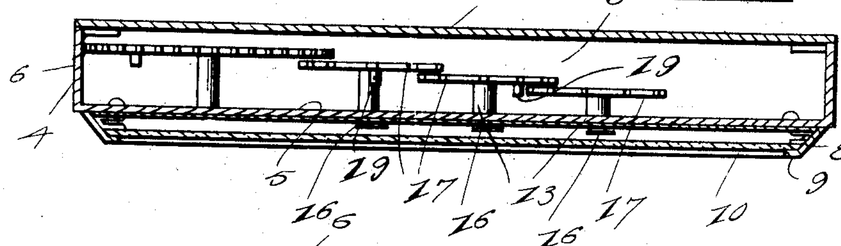
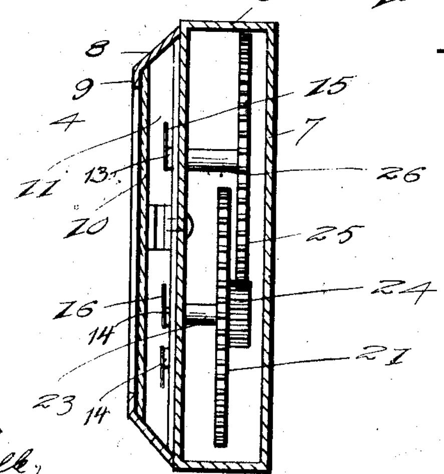



Fig. 5.

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# UNITED STATES PATENT OFFICE.

CHARLES M. HERSHEY, OF SHAMOKIN, PENNSYLVANIA.

COMPTOMETER.

**1,280,466.** Specification of Letters Patent. **Patented Oct. 1, 1918.**

Application filed November 17, 1916. Serial No. 131,857.

*To all whom it may concern:*

Be it known that I, CHARLES M. HERSHEY, a citizen of the United States, residing at Shamokin, in the county of Northumberland and State of Pennsylvania, have invented certain new and useful Improvements in Comptometers; and I do hereby declare the following to be a full, clear, and exact description of the invention, such as will enable others skilled in the art to which it appertains to make and use the same.

This invention relates to new and useful improvements in comptometers and the principal object of the invention is to provide a device which is adapted to be attached to a loom for computing the quantity of work performed by the operator and also simultaneously computing the wages of the operator.

Another object of the invention is to provide a device which is adapted to be attached to the crank shaft of a loom so that the number of revolutions of the crank shaft will be registered.

With these and other objects in view the invention consists in the novel combination and arrangement of parts which will be fully set forth in the following specification and accompanying drawings, in which:—

Figure 1 is a view showing the device applied to a loom.

Fig. 2 is a face view in elevation of the comptometer.

Fig. 3 is a rear view of the device showing the back cover plates removed.

Fig. 4 is a horizontal sectional view through the device.

Fig. 5 is a vertical sectional view through the apparatus.

Fig. 6 is a fragmentary view partly in section showing the device applied to a loom and illustrating in detail the method of supporting the same.

Referring to the drawings the numeral 1 designates a loom of the ordinary construction provided with the usual crank shaft 2 while the comptometer is designated generally by the numeral 3. This comptometer comprises the housing 4 consisting of the front wall 5, side and end walls 6 and the back wall 7 which is attached to the side and end walls in any suitable manner. The front wall is provided near its edges with forwardly and inwardly extending flanges 8 which are bent to provide the retaining flanges 9 by means of which a glass panel 10 is secured in place. These flanges and this panel coöperate in forming the dial chamber 11 in which the indicating hands move over the various dials designated by the numeral 12. The set of dials designated generally A are used for computing the wage of the operator of the machine while the dials designated by the character B are used for computing the quantity of work turned out and the number of revolutions made by the crank shaft of the machine. Contained within the housing 4 are the shafts 13 and 14 respectively on which the star wheels for operating the hands are mounted. These shafts are journaled in the front wall and extend through the same and are provided at their forward ends with the indicating hands 15 and 16 respectively which coöperate with the dials 12 in indicating the work and wage.

Mounted on the inner ends of the shafts 13 are star wheels 17 while the inner ends of the shafts 14 are provided with the star wheels 18. Suitable pins 19 are carried by the star wheels 17 while similar pins 20 are carried by the star wheels 18 and it will thus be seen that when one of the wheels makes a complete revolution the next adjacent wheel will be operated.

In order to drive the star wheel indicating the units on the work performed dials, a suitable gear 21 is provided which carries near its periphery a suitable pin 22 for engagement with the star wheel next adjacent to said gear. This gear is mounted on a stud 23 within the housing 4 and a pinion 24 is carried by said gear for meshing engagement with the gear 25 which is mounted on the stud 26 and carries a pin 27 for engagement with the star wheel 17 which indicates the unit. It will thus be seen that the speed of rotation of the gear wheels 25 will be considerably lower than that of the gear wheel 21. In order to properly drive the gear 21 a suitable worm 28 is mounted on a shaft 29 which is rotatably mounted in suitable bearings 30 and which shaft is connected to the crank shaft of the loom.

In order to properly time the device each of the gear wheels 21 and 25 is preferably provided with one hundred teeth. It will thus be seen that a single revolution of the shaft 29 will move the gear wheel 21 one-hundredth of a revolution. The pinion 24 is preferably provided with twenty-three teeth so that the gear 25 will rotate at a much slower rate of speed than the gear 21. It will be understood however, that the number of teeth of the pinion 24 may be varied to meet various requirements so that the wage may be varied per number of picks.

It will be apparent from the foregoing that in use the shaft 29 rotates according to the speed of rotation of the crank shaft of the loom and such rotation causes the gears 21 and 25 to rotate thereby causing their pins 22 and 27 to engage the star wheels 18 and 17 respectively thus moving said star wheels a partial revolution.

As shown in the drawings each of the star wheels is provided with ten teeth and one of said teeth carries a stud adapted to engage the next adjacent star wheel. In this way it will be seen that the gear 21 must make ten revolutions before the star wheel next adjacent thereto has made a single revolution and the star wheel must make ten revolutions to move the next adjacent star wheel. Thus it will be seen that the rate will be calculated according to tens.

In order to prevent the star wheels from moving beyond a tenth revolution each time it is engaged by one of the studs suitable slotted posts 30 are attached to the wall 5 and extend inwardly therefrom and fitted in the slots in said posts are springs 31, the ends of which are curved as at 32 and adapted to engage in the notches between the forks of the star wheels. It will thus be seen that the springs will tend to hold the star wheels against revolution and yet will be sufficiently resilient to allow the points of said wheels to pass when the wheels are moved under the influence of the pins.

The crank shaft 2 is driven from the shaft $2^a$ through the medium of beveled gears $2^b$ and $2^c$. The gear $2^c$ is provided with sockets $2^d$ for the reception of the prongs or fingers $2^e$ of a coupling $2^f$ through the medium of which the comptometer shaft 29 is connected in axial alinement with the crank shaft 2. The comptometer 3 is secured to the loom to position and hold its shaft 29 in axial alinement with the crank shaft 2 by a bracket $2^g$.

While in the foregoing there has been shown and described the preferred embodiment of this invention, it is to be understood that such changes may be made in the combination and arrangement of parts as will fall within the spirit and scope of the invention as claimed.

What is claimed is:—

A loom actuated pick counting and wage calculating device, comprising two series of dials having indicator hands, a rotatable gear wheel, a shaft adapted to be driven by a rotating part of the loom, a worm fixed to said shaft and meshing with said gear wheel, a train of star wheels adapted to be driven by said gear wheel and adapted to rotate one series of said indicator hands to register the number of picks of the loom, a changeable pinion attached to said gear wheel, a second gear wheel meshing with said pinion, and a second train of star wheels adapted to be driven by said second gear wheel and adapted to operate the other series of indicator hands to register the wages due the operator of the loom.

In testimony whereof I affix my signature in presence of two witnesses.

CHARLES M. HERSHEY.

Witnesses:

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